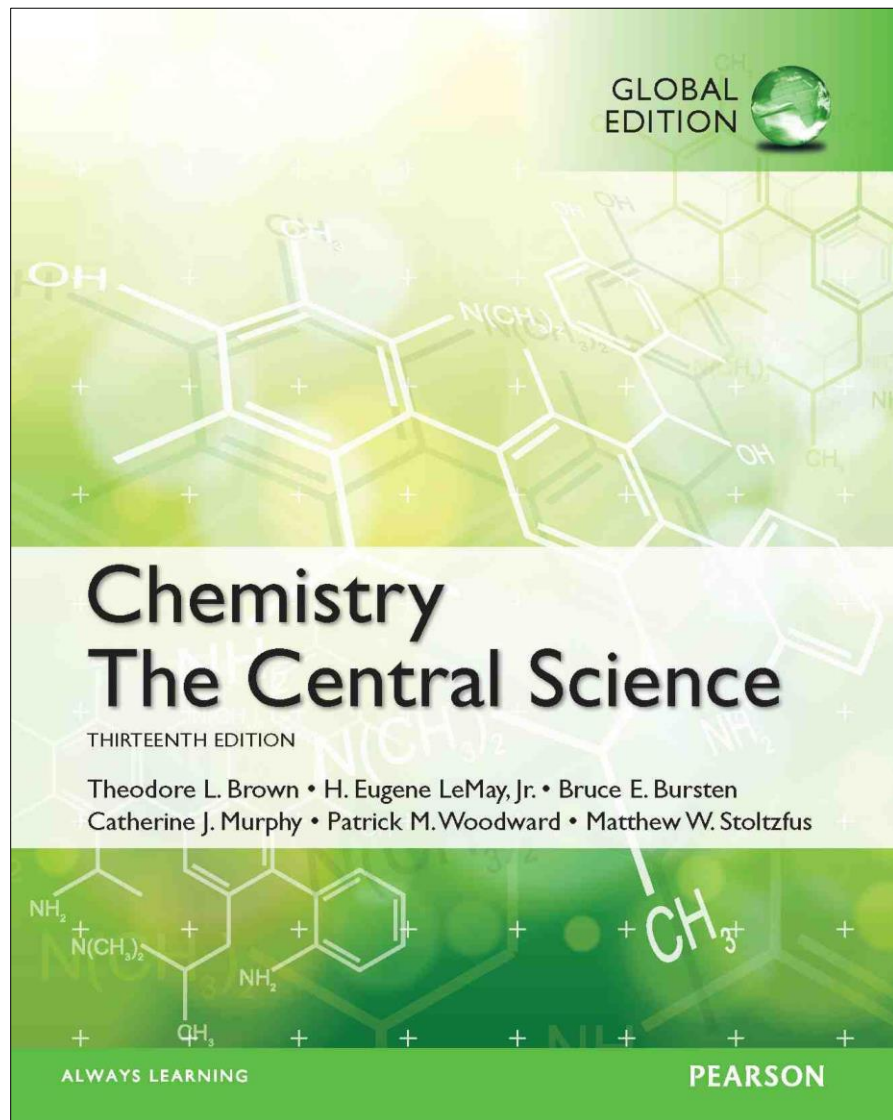




Lecture Presentation

Chapter 19

Chemical Thermodynamics

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Hamden, CT

First Law of Thermodynamics

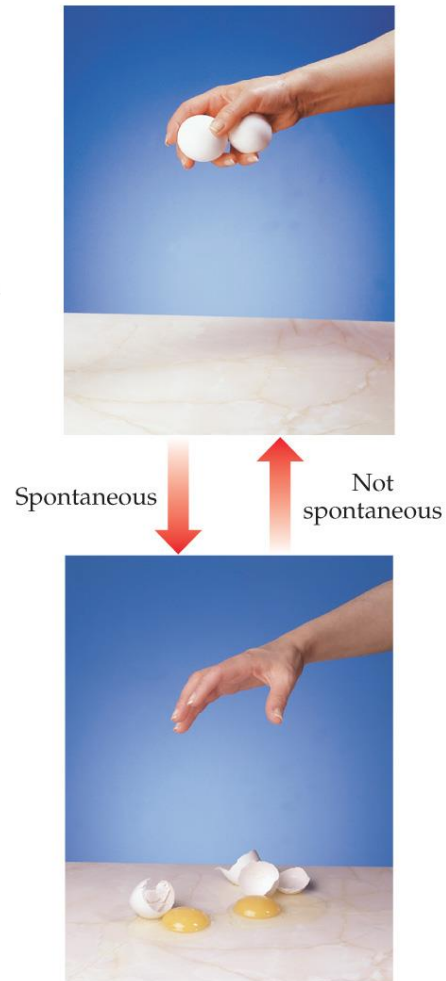
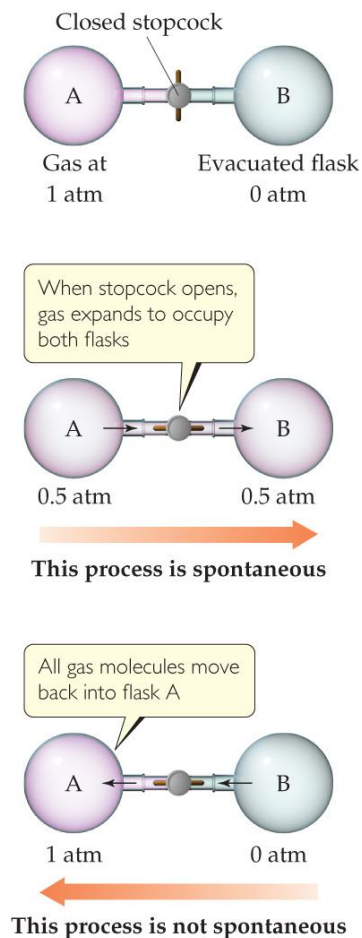
- You will recall from Chapter 5 that energy cannot be created or destroyed.
- Therefore, the total energy of the universe is a constant.
- Energy can, however, be converted from one form to another or transferred from a system to the surroundings or vice versa.

Enthalpy/Entropy

- Change in **enthalpy** is the heat absorbed by a system during a **constant-pressure** process.
- **Entropy** is a measure of the randomness in a system.
- Both play a role in determining whether a process is spontaneous.

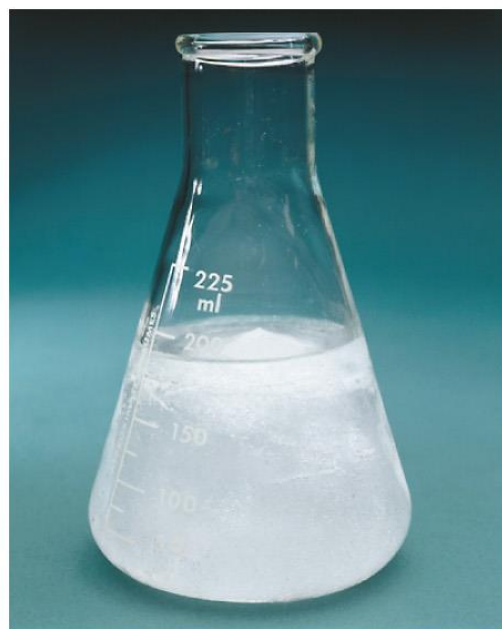
Spontaneous Processes

- **Spontaneous processes** proceed without any outside assistance.
- The gas in vessel A will spontaneously effuse into vessel B, but it will *not* spontaneously return to vessel A.
- Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.



Experimental Factors Affect Spontaneous Processes

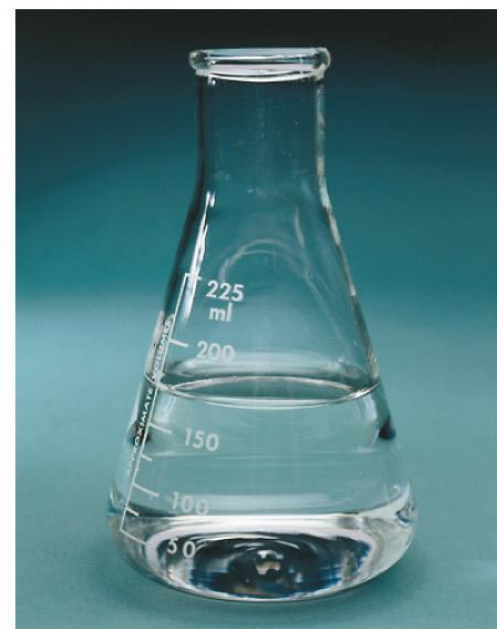
- **Temperature** and **pressure** can affect spontaneity.
- An example of how temperature affects spontaneity is ice melting or freezing.



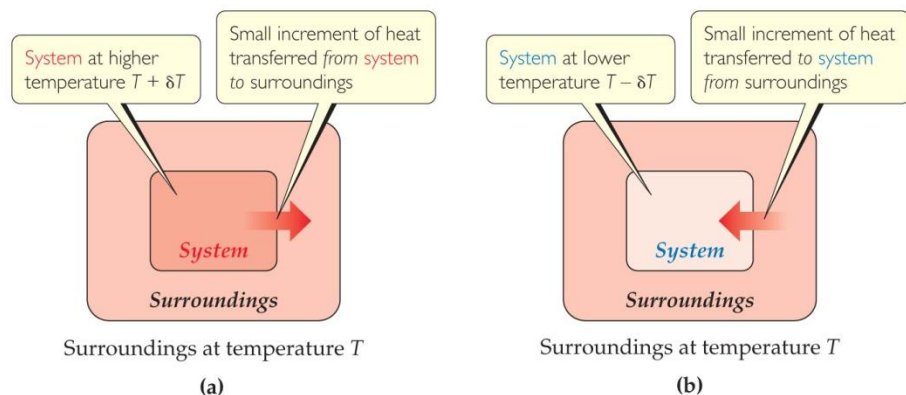
Spontaneous for $T > 0\text{ }^{\circ}\text{C}$



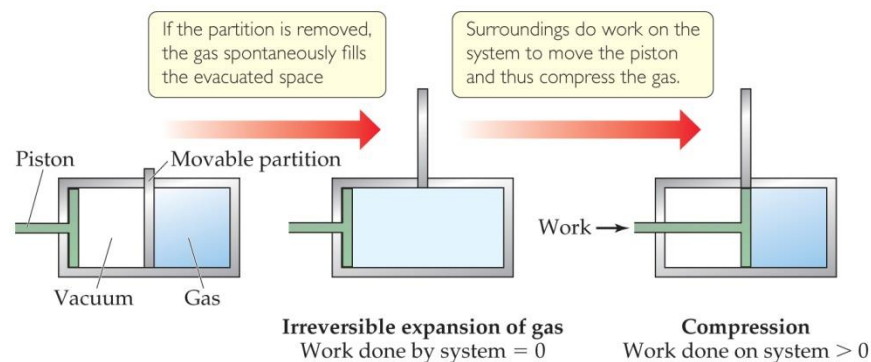
Spontaneous for $T < 0\text{ }^{\circ}\text{C}$



Reversible and Irreversible Processes



Reversible process: The system changes so that the system and surroundings can **be returned to the original state by exactly reversing the process..**



Irreversible processes cannot be undone by exactly reversing the change to the system or cannot have the process exactly followed in reverse. Also, **any spontaneous process is irreversible!**

Entropy

- **Entropy** can be thought of as a measure of the randomness of a system.
- It is a state function:

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

For the special case of an isothermal process, ΔS is equal to the heat that would be transferred if the process were reversible, q_{rev} , divided by the absolute temperature at which the process occurs:

$$\Delta S = \frac{q_{\text{rev}}}{T} \quad (\text{constant } T) \quad [19.2]$$

Although many possible paths can take the system from one state to another, only one path is associated with a reversible process. Thus, the value of q_{rev} is uniquely defined for any two states of the system. Because S is a state function, we can use Equation 19.2 to calculate ΔS for *any* isothermal process between states, not just the reversible one.

ΔS for Phase Changes

The enthalpy of fusion for H_2O is $\Delta H_{\text{fusion}} = 6.01 \text{ kJ/mol}$ (a positive value because melting is an endothermic process). Thus, we can use Equation 19.2 to calculate ΔS_{fusion} for melting 1 mol of ice at 273 K:

$$\Delta S_{\text{fusion}} = \frac{q_{\text{rev}}}{T} = \frac{\Delta H_{\text{fusion}}}{T} = \frac{(1 \text{ mol})(6.01 \times 10^3 \text{ J/mol})}{273 \text{ K}} = 22.0 \text{ J/K}$$

SAMPLE EXERCISE 19.2

Calculating ΔS for a Phase Change

Elemental mercury is a silver liquid at room temperature. Its normal freezing point is $-38.9\text{ }^{\circ}\text{C}$, and its molar enthalpy of fusion is $\Delta H_{\text{fusion}} = 2.29\text{ kJ/mol}$. What is the entropy change of the system when 50.0 g of $\text{Hg}(l)$ freezes at the normal freezing point?

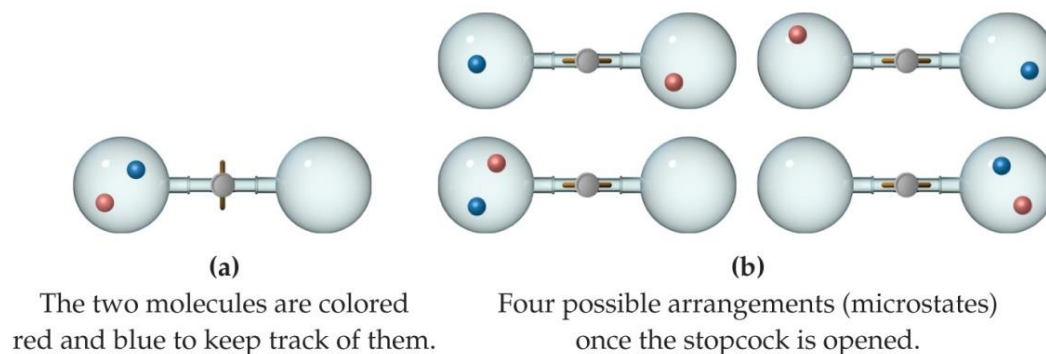
Second Law of Thermodynamics

- The entropy of the universe increases in any spontaneous processes.
- This results in the following relationships:

Reversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$

Irreversible Process: $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$

Entropy on the Molecular Scale



- Boltzmann described entropy on the molecular level.
- Gas molecule expansion: Two molecules are in the apparatus above; both start in one side. What is the likelihood they both will end up there? $(1/2)^2$
- If one mole is used? $(1/2)^{6.02 \times 10^{23}}$! (No chance!)
- Gases spontaneously expand to fill the volume given.
- Most probable arrangement of molecules: approximately equal molecules in each side

Statistical Thermodynamics

- Thermodynamics looks at **bulk** properties of substances (the *big* picture).
- We have seen what happens on the **molecular scale**.
- How do they relate?
- We use statistics (probability) to relate them. The field is called statistical thermodynamics.
- Microstate: A single possible arrangement of **position** and **kinetic energy** of molecules

Boltzmann's Use of Microstates

- Because there are so many possible microstates, we can't look at every picture.
- W represents the number of microstates.
- **Entropy** is a measure of how many microstates are associated with a particular macroscopic state.
- The connection between the number of microstates and the entropy of the system is:

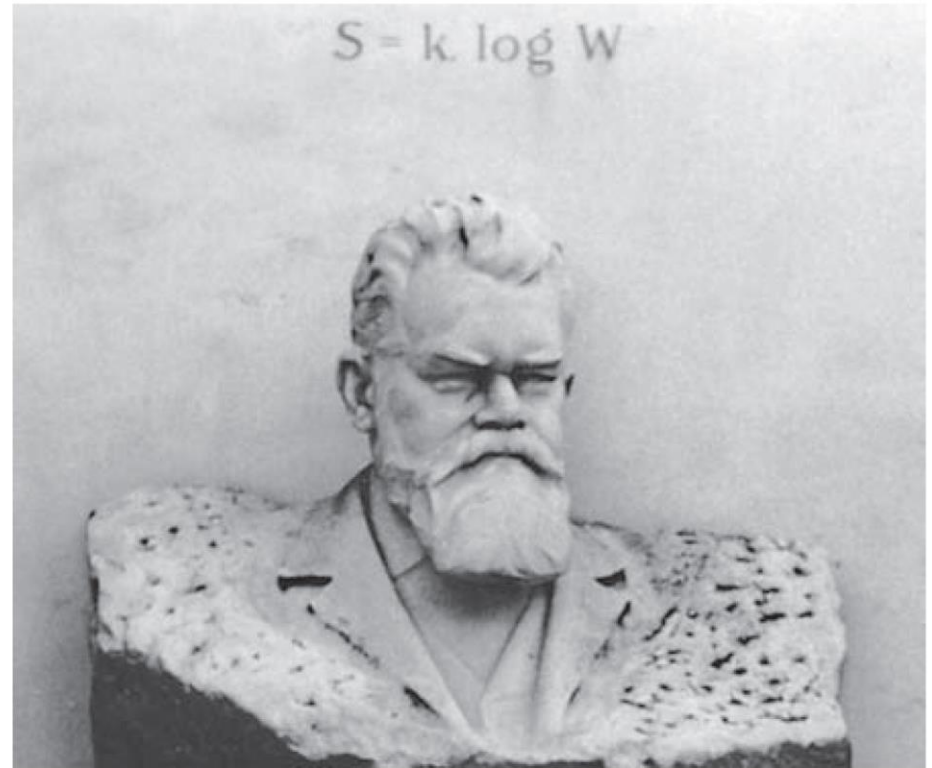
$$S = k \ln W$$

In this equation, k is the Boltzmann constant, 1.38×10^{-23} J/K. Thus, *entropy is a measure of how many microstates are associated with a particular macroscopic state.*

Boltzmann's Use of Microstates

- The connection between the number of microstates and the entropy of the system

$$S = k \ln W$$



▲ **Figure 19.7** Ludwig Boltzmann's gravestone.

Boltzmann's gravestone in Vienna is inscribed with his famous relationship between the entropy of a state, S , and the number of available microstates, W . (In Boltzmann's time, "log" was used to represent the natural logarithm.)

Entropy Change

- Since entropy is a **state function**, the final value minus the initial value will give the overall change.
- In this case, an increase in the number of microstates results in a positive entropy change (more disorder).

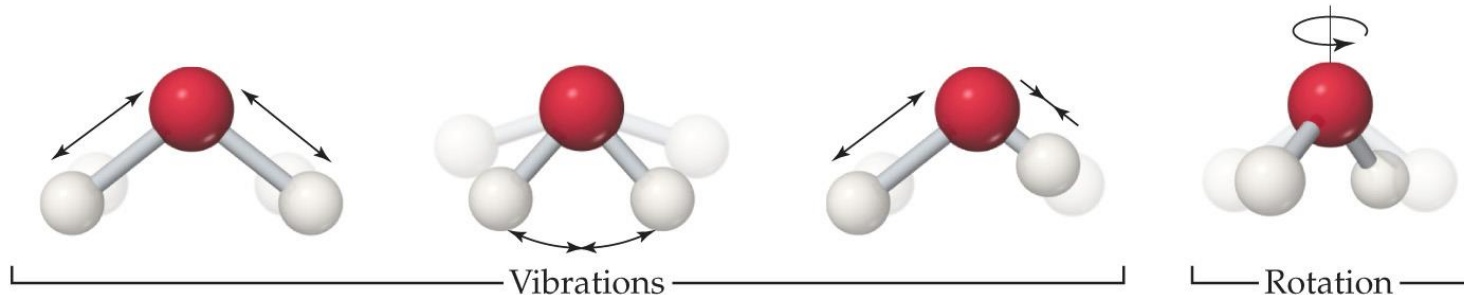
$$\Delta S = k \ln W_{\text{final}} - k \ln W_{\text{initial}} = k \ln \frac{W_{\text{final}}}{W_{\text{initial}}}$$

Effect of Volume and Temperature Change on the System

- If we increase **volume**, there are more **positions** possible for the molecules. This results in more microstates, so increased entropy.
- If we increase **temperature**, the **average kinetic energy increases**. This results in a greater distribution of molecular speeds. Therefore, there are more possible kinetic energy values, resulting in more microstates, increasing entropy.

Molecular Motions

- Molecules exhibit several types of motion.
 - **Translational**: Movement of the entire molecule from one place to another
 - **Vibrational**: Periodic motion of atoms within a molecule (change in bond lengths or angles)
 - **Rotational**: Rotation of the molecule about an axis
- Note: More *atoms* means more microstates (more possible molecular motions).

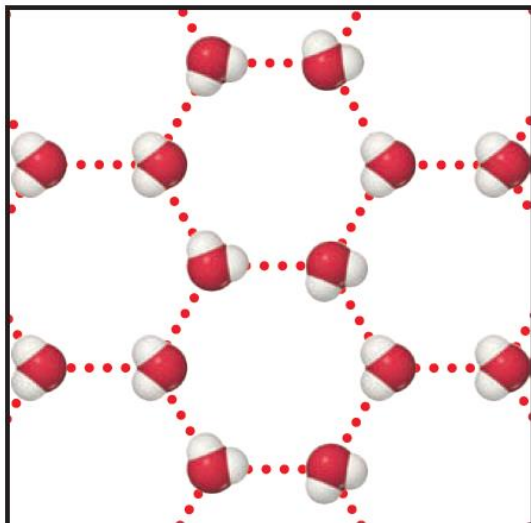


In which phase are water molecules least able to have rotational motion?

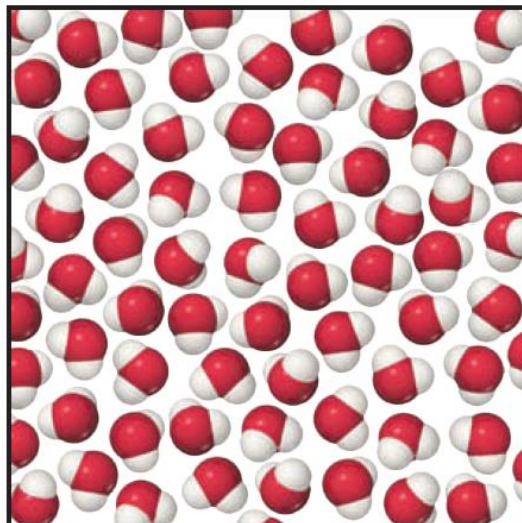
Increasing entropy



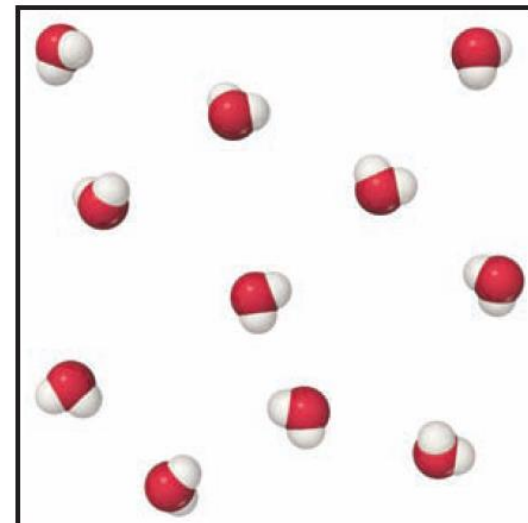
Ice



Liquid water



Water vapor



Rigid, crystalline structure
Motion restricted to **vibration** only
Smallest number of microstates

Increased freedom with respect
to **translation**
Free to **vibrate** and **rotate**
Larger number of microstates

Molecules spread out, essentially
independent of one another
Complete freedom for **translation**,
vibration, and **rotation**
Largest number of microstates

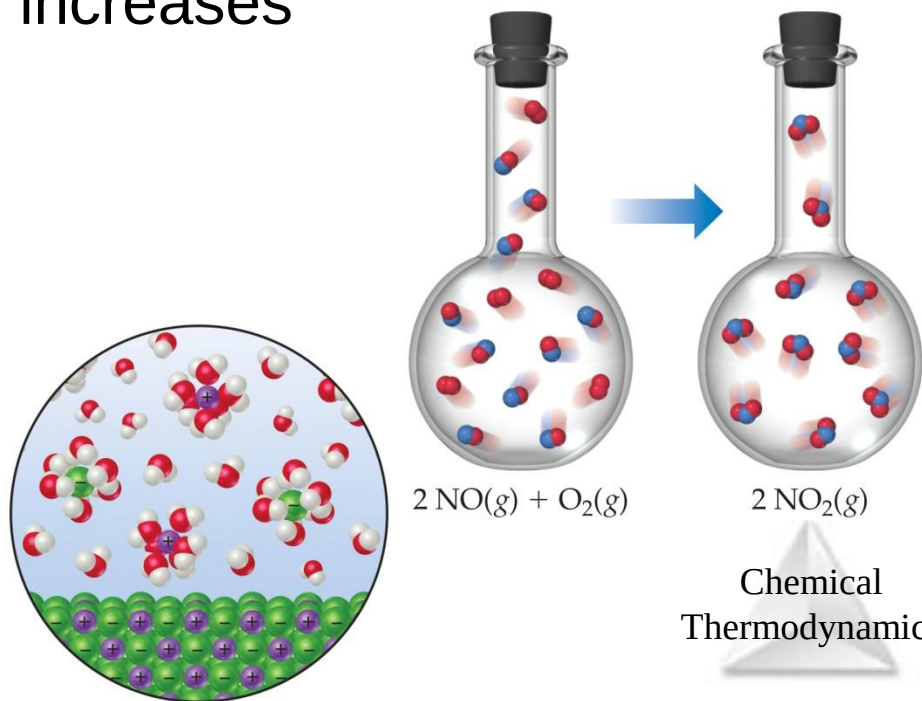
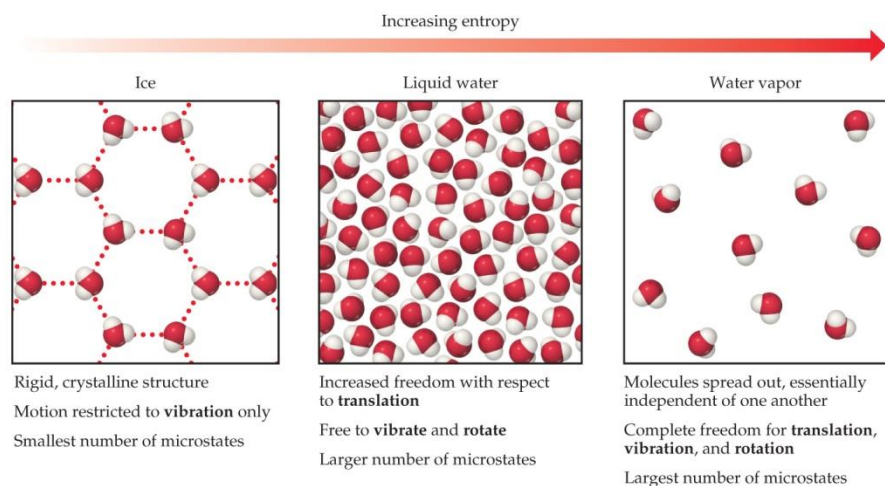
Entropy on the Molecular Scale

The number of microstates and, therefore, the entropy tend to increase with increases in

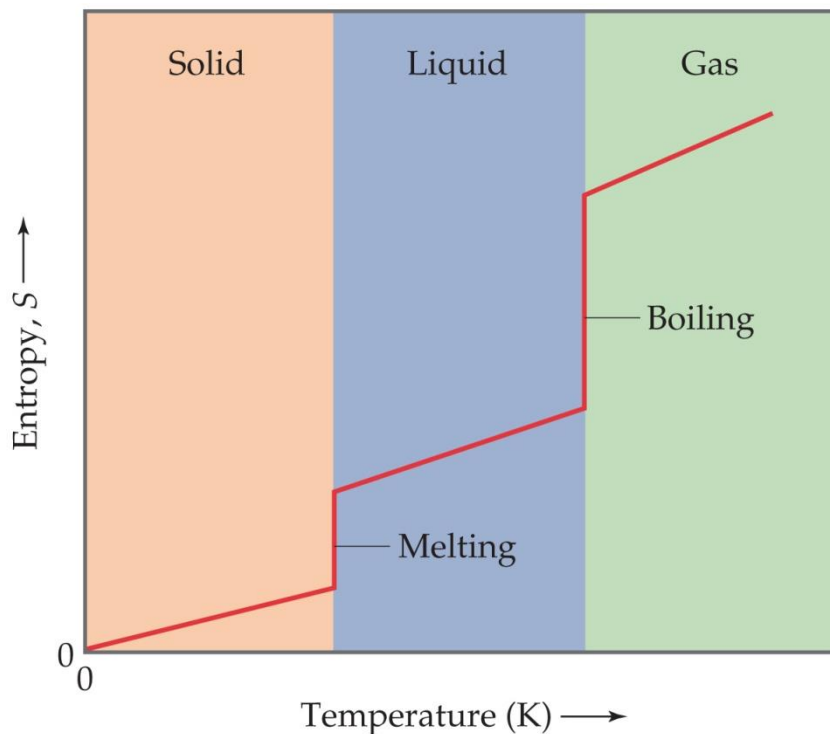
- temperature.
- volume.
- the number of independently moving molecules.

Entropy and Physical States

- Entropy increases with the freedom of motion of molecules.
- $S(g) > S(l) > S(s)$
- Entropy of a system increases for processes where
 - gases form from either solids or liquids.
 - liquids or solutions form from solids.
 - the number of gas molecules increases during a chemical reaction.



Third Law of Thermodynamics



- The entropy of a pure crystalline substance at absolute zero is 0.
- Consider all atoms or molecules in the perfect lattice at 0 K; there will only be *one* microstate.
- $S = k \ln W = k \ln 1 = 0$

Standard Entropies

- The reference for entropy is 0 K, so the values for elements are *not* 0 J/mol K at 298 K.
- Standard molar enthalpy for gases are generally greater than liquids and solids. (Be careful of size!)
- Standard entropies increase with molar mass.
- Standard entropies increase with number of atoms in a formula.

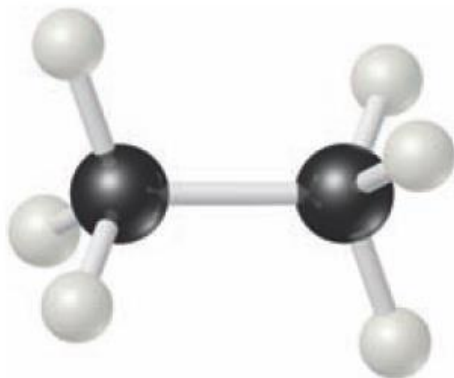
Table 19.1 Standard Molar Entropies of Selected Substances at 298 K

Substance	S° (J/mol·K)
$\text{H}_2(\text{g})$	130.6
$\text{N}_2(\text{g})$	191.5
$\text{O}_2(\text{g})$	205.0
$\text{H}_2\text{O}(\text{g})$	188.8
$\text{NH}_3(\text{g})$	192.5
$\text{CH}_3\text{OH}(\text{g})$	237.6
$\text{C}_6\text{H}_6(\text{g})$	269.2
$\text{H}_2\text{O}(\text{l})$	69.9
$\text{CH}_3\text{OH}(\text{l})$	126.8
$\text{C}_6\text{H}_6(\text{l})$	172.8
$\text{Li}(\text{s})$	29.1
$\text{Na}(\text{s})$	51.4
$\text{K}(\text{s})$	64.7
$\text{Fe}(\text{s})$	27.23
$\text{FeCl}_3(\text{s})$	142.3
$\text{NaCl}(\text{s})$	72.3

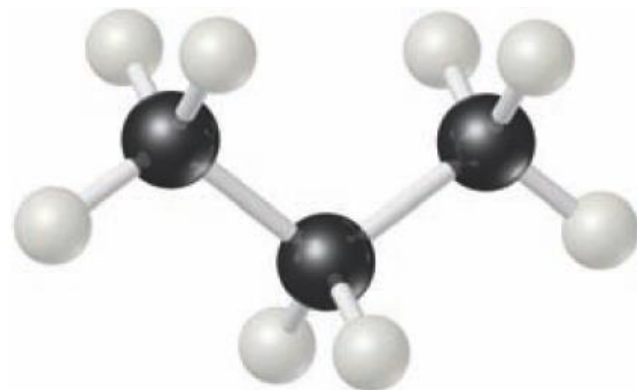
Standard Entropies



Methane, CH₄
 $S^\circ = 186.3 \text{ J/mol-K}$



Ethane, C₂H₆
 $S^\circ = 229.6 \text{ J/mol-K}$



Propane, C₃H₈
 $S^\circ = 270.3 \text{ J/mol-K}$

▲ **Figure 19.13** Entropy increases with increasing molecular complexity.

Entropy Changes

Entropy changes for a reaction can be calculated in a manner analogous to that by which ΔH is calculated:

$$\Delta S^{\circ} = \sum n S^{\circ} \text{ (products)} - \sum m S^{\circ} \text{ (reactants)}$$

where n and m are the coefficients in the balanced chemical equation.

SAMPLE EXERCISE 19.5

Calculating ΔS° from Tabulated Entropies

Calculate the change in the standard entropy of the system, ΔS° , for the synthesis of ammonia from $\text{N}_2(g)$ and $\text{H}_2(g)$ at 298 K:

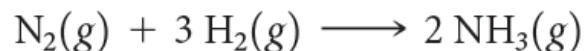


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$\text{CH}_3\text{OH}(g)$	237.6
$\text{C}_6\text{H}_6(g)$	269.2
$\text{H}_2\text{O}(l)$	69.9
$\text{CH}_3\text{OH}(l)$	126.8
$\text{C}_6\text{H}_6(l)$	172.8
$\text{Li}(s)$	29.1
$\text{Na}(s)$	51.4
$\text{K}(s)$	64.7
$\text{Fe}(s)$	27.23
$\text{FeCl}_3(s)$	142.3
$\text{NaCl}(s)$	72.3

Entropy Changes in Surroundings

- Heat that flows into or out of the system changes the entropy of the surroundings.
- For an isothermal process

$$\Delta S_{\text{surr}} = \frac{-q_{\text{sys}}}{T}$$

- At **constant pressure**, q_{sys} is simply ΔH° for the system.

$$\Delta S_{\text{surr}} = \frac{-\Delta H_{\text{sys}}}{T} \quad [\text{at constant } P]$$

Entropy Change in the Universe

- The universe is composed of the system and the surroundings.

- Therefore,

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

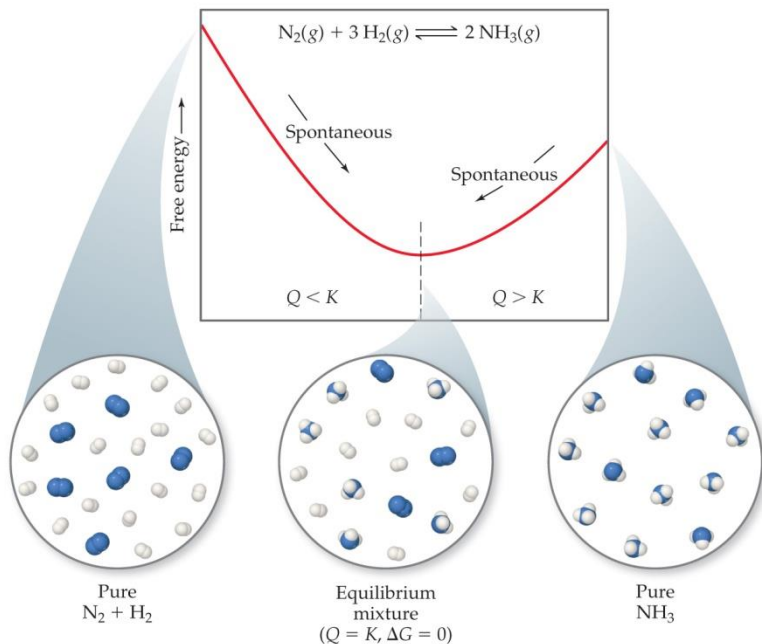
- For spontaneous processes

$$\Delta S_{\text{universe}} > 0$$

Total Entropy and Spontaneity

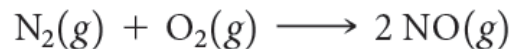
- $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$
- Substitute for the entropy of the surroundings:
 - $\Delta S_{\text{universe}} = \Delta S_{\text{system}} - \Delta H_{\text{system}}/T$
- Multiply by $-T$:
 - $-T\Delta S_{\text{universe}} = -T\Delta S_{\text{system}} + \Delta H_{\text{system}}$
- Rearrange:
 - $-T\Delta S_{\text{universe}} = \Delta H_{\text{system}} - T\Delta S_{\text{system}} = \Delta(H_{\text{system}} - TS_{\text{system}})$
- $G = H - TS$, the **Gibbs Free Energy** - a new state function
and $\Delta G = \Delta H - T\Delta S$ (*isothermal*)

Gibbs Free Energy



1. If ΔG is negative, the forward reaction is spontaneous ($\Delta s_{\text{universe}}$ **+**).
2. If ΔG is 0, the system is at equilibrium ($\Delta s_{\text{universe}}$ **0**).
3. If ΔG is positive ($\Delta s_{\text{universe}}$ **-**), the reaction is spontaneous in the reverse direction.

Calculate the standard free-energy change for the formation of $\text{NO}(g)$ from $\text{N}_2(g)$ and $\text{O}_2(g)$ at 298 K:



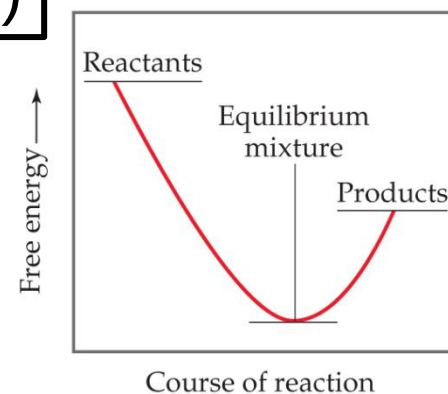
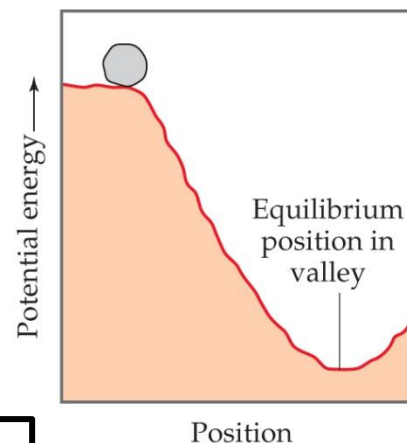
given that $\Delta H^\circ = 180.7 \text{ kJ}$ and $\Delta S^\circ = 24.7 \text{ J/K}$. Is the reaction spontaneous under these conditions?

Standard Free Energy Changes

Standard free energies of formation ΔG_f° are analogous to standard enthalpies of formation (ΔH_f°)

$$\Delta G_{rxn}^\circ = \sum n \Delta G_f^\circ (\text{products}) - \sum m \Delta G_f^\circ (\text{reactants})$$

where n and m are the stoichiometric coefficients.



Chemical
Thermodynamics

Free Energy Changes

Table 19.3 How Signs of ΔH and ΔS Affect Reaction Spontaneity

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
–	+	–	–	Spontaneous at all temperatures	$2 \text{O}_3(g) \longrightarrow 3 \text{O}_2(g)$
+	–	+	+	Nonspontaneous at all temperatures	$3 \text{O}_2(g) \longrightarrow 2 \text{O}_3(g)$
–	–	+	+ or –	Spontaneous at low T ; nonspontaneous at high T	$\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(s)$
+	+	–	+ or –	Spontaneous at high T ; nonspontaneous at low T	$\text{H}_2\text{O}(s) \longrightarrow \text{H}_2\text{O}(l)$

- How does ΔG change with temperature?
- $\Delta G = \Delta H - T\Delta S$
- Since reactions are spontaneous if $\Delta G < 0$, the sign of enthalpy and entropy and the magnitude of the temperature matters to spontaneity.

Standard Free Energy

**Table 19.2 Conventions Used
in Establishing Standard Free
Energies**

State of Matter	Standard State
Solid	Pure solid
Liquid	Pure liquid
Gas	1 atm pressure
Solution	1 <i>M</i> concentration
Element	$\Delta G_f^\circ = 0$ for element in standard state

Standard Enthalpies of Formation

By definition, the standard enthalpy of formation of the most stable form of any element is zero because there is no formation reaction needed when the element is already in its standard state.

Thus, the values of ΔH_f° for **C(graphite)**, **H₂(g)**, **O₂(g)**, and the standard states of other elements are zero by definition.

See Table 5.3 and Appendix C for listed ΔH_f° data.

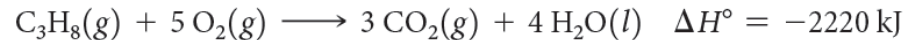
SAMPLE EXERCISE 19.7

Calculating Standard Free-Energy Change from Free Energies of Formation

(a) Use data from Appendix C to calculate the standard free-energy change for the reaction $\text{P}_4(\text{g}) + 6 \text{Cl}_2(\text{g}) \longrightarrow 4 \text{PCl}_3(\text{g})$ at 298 K. (b) What is ΔG° for the reverse of this reaction?

SAMPLE EXERCISE 19.8 Predicting and Calculating ΔG°

In Section 5.7 we used Hess's law to calculate ΔH° for the combustion of propane gas at 298 K:



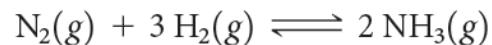
(a) *Without using data from Appendix C*, predict whether ΔG° for this reaction is more negative or less negative than ΔH° . (b) Use data from Appendix C to calculate ΔG° for the reaction at 298 K. Is your prediction from part (a) correct?

SAMPLE

EXERCISE 19.9

Determining the Effect of Temperature on Spontaneity

The Haber process for the production of ammonia involves the equilibrium



Assume that ΔH° and ΔS° for this reaction do not change with temperature. **(a)** Predict the direction in which ΔG for the reaction changes with increasing temperature. **(b)** Calculate ΔG at 25 °C and at 500 °C.

Free Energy and Equilibrium

Under **any conditions**, standard or nonstandard, the free energy change can be found this way:

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

(Under **standard conditions**, concentrations are 1 *M*, so $Q = 1$ and $\ln Q = 0$; the last term drops out.)

Free Energy and Equilibrium

- At equilibrium, $Q = K$, and $\Delta G = 0$.
- The equation becomes

$$0 = \Delta G^{\circ} + RT \ln K$$

- Rearranging, this becomes

$$\Delta G^{\circ} = -RT \ln K$$

or

$$K = e^{-\Delta G^{\circ}/RT}$$

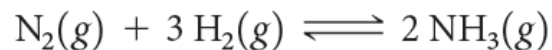
SAMPLE

EXERCISE 19.10 Relating ΔG to a Phase Change at Equilibrium

(a) Write the chemical equation that defines the normal boiling point of liquid carbon tetrachloride, $\text{CCl}_4(l)$. (b) What is the value of ΔG° for the equilibrium in part (a)? (c) Use data from Appendix C and Equation 19.12 to estimate the normal boiling point of CCl_4 .

**SAMPLE
EXERCISE 19.11****Calculating the Free-Energy Change under
Nonstandard Conditions**

Calculate ΔG at 298 K for a mixture of 1.0 atm N_2 , 3.0 atm H_2 , and 0.50 atm NH_3 being used in the Haber process:



SAMPLE EXERCISE 19.12 Calculating an Equilibrium Constant from ΔG°

The standard free-energy change for the Haber process at 25 °C was obtained in Sample Exercise 19.9 for the Haber reaction:



Use this value of ΔG° to calculate the equilibrium constant for the process at 25 °C.